



## Technical Debt and DevOps: Strategies for Managing Legacy Systems in a CI/CD World

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### Abstract

Legacy systems continue to form the backbone of critical business operations in sectors such as finance, healthcare, and manufacturing. However, their monolithic architectures, outdated technologies, and lack of modularity create significant friction when organizations attempt to adopt modern DevOps practices and continuous integration/continuous delivery (CI/CD) workflows. This research paper, titled *“Technical Debt and DevOps: Strategies for Managing Legacy Systems in a CI/CD World,”* investigates the intersection of technical debt and DevOps implementation, with a particular focus on managing and modernizing legacy systems without disrupting delivery pipelines.

The core objective of the study is to analyze how accumulated technical debt—stemming from outdated codebases, undocumented APIs, and fragile dependencies—undermines the agility and reliability of CI/CD processes. The paper identifies common manifestations of technical debt in legacy environments, such as poor testability, low cohesion, and high code churn, and correlates these with key DevOps metrics including deployment frequency, mean time to recovery (MTTR), and change failure rate. Through case studies and tool-based assessments, this research explores real-world strategies for mitigating these challenges.

Specifically, the study evaluates approaches such as the strangler fig pattern for gradual modernization, automated regression testing to secure legacy behaviors, static code analysis tools (e.g., SonarQube) to quantify debt, and containerization to abstract legacy runtimes. It also examines the use of infrastructure-as-code (IaC), blue-green deployments, and continuous observability to insulate CI/CD pipelines from the instability often introduced by legacy code. The paper further introduces a technical debt remediation matrix to prioritize refactoring efforts based on business value and architectural risk.

Findings indicate that while complete legacy rewrites are impractical for most enterprises, incremental debt repayment combined with targeted DevOps integration can yield substantial improvements in delivery velocity and quality. The study emphasizes the need for cross-functional collaboration among developers, testers, and platform engineers to institutionalize practices that prevent new debt accumulation while managing existing liabilities. Ultimately, the research contributes a structured roadmap for organizations striving to reconcile the operational inertia of legacy systems with the rapid, iterative nature of modern software delivery.



**Keywords :** Legacy systems , critical , business operations , finance, healthcare, and manufacturing

## Introduction

The landscape of modern software engineering is increasingly defined by the need for agility, rapid iteration, and continuous value delivery. In this context, DevOps—a cultural and technical movement that emphasizes collaboration, automation, and continuous feedback between development and operations—has emerged as a critical enabler of high-velocity software delivery. DevOps practices, supported by CI/CD (Continuous Integration/Continuous Delivery) pipelines, allow teams to release new features, updates, and bug fixes in a seamless, reliable, and reproducible manner. These workflows have become the norm in greenfield projects and cloud-native environments, offering a powerful solution to the challenges of traditional waterfall or manual deployment models.

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However, as organizations seek to modernize their software delivery practices, many are confronted with the persistent reality of legacy systems—software that is essential for daily operations but was developed under outdated paradigms and constraints. These legacy systems, often written in monolithic styles, rely on obsolete frameworks, lack automated test coverage, and are tightly coupled with physical infrastructure. While replacing such systems entirely may be ideal in theory, the operational risks, high costs, and business dependencies often make full-scale rewrites infeasible. This dilemma places enterprises at a crossroads: how can they integrate modern DevOps methodologies into environments weighed down by the inertia of the past?

A key concept that underpins this challenge is technical debt. Coined by Ward Cunningham, technical debt refers to the implied cost of additional rework caused by choosing suboptimal or expedient solutions in software design and development. In the context of legacy systems, technical debt is not only a metaphor but a tangible barrier to progress. It accumulates in the form of outdated codebases, poor modularity, hardcoded dependencies, lack of documentation, brittle interfaces, and fragile deployment processes. As organizations strive to implement automated testing, continuous delivery, and infrastructure-as-code, they often find that existing debt inhibits pipeline stability, increases change failure rates, and slows down iteration cycles.

The confluence of DevOps transformation and technical debt management is especially pronounced in regulated industries such as finance, insurance, and healthcare, where compliance requirements and risk sensitivity further limit the ability to refactor or decommission legacy systems. These environments typically operate mission-critical software developed over decades, often by teams that have since disbanded. The absence of maintainers, along with institutional knowledge silos, complicates efforts to introduce modern practices without breaking existing functionality. Moreover, tools and workflows designed for cloud-native applications may not be readily applicable to tightly coupled legacy applications that were never designed with automation or modularity in mind.

The motivation for this research stems from the urgent need to bridge the gap between legacy software and modern delivery expectations. As businesses increasingly rely on digital products and services to maintain competitiveness, the pressure to innovate faster without sacrificing reliability is mounting. Yet, for many organizations, technical debt has become a



bottleneck that prevents them from scaling DevOps practices effectively. The core question becomes: how can enterprises manage and mitigate technical debt in legacy systems while adopting DevOps and CI/CD principles in a sustainable and secure way?

To address this question, this paper explores a comprehensive framework of strategies, patterns, and tools aimed at enabling DevOps adoption in legacy-heavy environments. These include the use of microservice decomposition, containerization, test automation, static analysis, progressive delivery, and observability tooling, all tailored to address specific forms of technical debt. The paper also examines organizational factors such as change management, cross-team collaboration, and stakeholder alignment, recognizing that technical debt is as much a socio-technical issue as it is a coding problem.

By focusing on the integration of DevOps practices within legacy system contexts, this research contributes to both academic understanding and practical application. It offers a roadmap for decision-makers and engineers who must balance the twin demands of reliability and innovation, and who seek to evolve their legacy ecosystems into more agile, maintainable, and delivery-capable platforms. In doing so, it affirms that technical debt, while inevitable, is not unmanageable—particularly when approached with discipline, tooling, and a clear architectural vision aligned with DevOps principles.

Ultimately, this paper underscores that technical debt management is no longer a secondary concern in the age of continuous delivery—it is a foundational challenge that must be strategically addressed if enterprises are to reap the full benefits of DevOps at scale.

## Literature Review

### 3.1 Foundations of Technical Debt Research

Ward Cunningham’s metaphor of *technical debt*—the interest a team “pays” each time an expedient design decision slows future change—moved from anecdote to measurable software-engineering construct in the mid-2010s. Work presented at the first ACM/IEEE **International Conference on Technical Debt (TechDebt 2018)** systematised vocabulary (code, architecture, test, requirements debt) and promoted tool-assisted quantification via static analysis and repository mining ([dl.acm.org](https://dl.acm.org)). Empirical studies soon confirmed the business impact: high defect density, increased mean time-to-recovery (MTTR) and lower deployment frequency in debt-laden components. Lenarduzzi *et al.* found that 38 % of build failures in a Finnish SME were traceable to files with elevated SonarQube “debt ratio” scores, underscoring the linkage between automated metrics and delivery risk (2017). Collectively, this early body of work framed debt not merely as code smell but as a socio-technical risk requiring continuous governance.

### 3.2 Quantifying and Prioritising Debt in CI Environments

Quantification research from 2016–2019 focused on aligning static-analysis output with the cadence of **Continuous Integration (CI)**. The *Technical Debt Dataset* (TDD) released in 2019 mined 30 open-source projects across five years, pairing SonarQube issues with commit metadata to expose “debt diffusion” hotspots—files that repeatedly attract fixes yet remain highly coupled ([dl.acm.org](https://dl.acm.org)). Li *et al.* tested the accuracy of SonarQube remediation-time estimates and showed a median error of just  $\pm 12\%$ , validating their use in sprint planning



([sciencedirect.com](https://www.sciencedirect.com)). Nevertheless, teams struggle to convert raw findings into backlog items. A 2018 exploratory case study on four companies reported that engineers prioritised only 14 % of detected issues, favouring defects that threatened release deadlines over architectural debt with long-term repercussions ([researchgate.net](https://www.researchgate.net)). These studies highlight the need for debt metrics that integrate seamlessly with pipeline dashboards and business value calculations.

### 3.3 Legacy Systems and the DevOps Adoption Paradox

Legacy applications—monolithic, tightly coupled, and often hosted on physical servers—constitute the bulk of enterprise portfolios in finance, telecom, and healthcare. Agile Alliance practitioners described how *Stop-the-Line* exercises exposed fragile batch scripts and manual deployment steps that acted as “interest payments” every sprint ([agilealliance.org](https://www.agilealliance.org)). Chef’s 2019 white-paper “**Continuous Delivery: Stop Making Exceptions**” echoed this view, noting that 70 % of exception paths in deployment run-books served only to appease ageing systems with hard-coded environment assumptions ([chef.io](https://www.chef.io)). Technical debt here manifests as brittle integration tests, missing artefact provenance, and inconsistent rollback procedures—each impeding the fast feedback loops central to DevOps.

### 3.4 Modernisation Pathways: Microservices & Strangler Patterns

Systematic mapping work by Alves *et al.* (2020) identified six macro-activities—assessment, decomposition, refactoring, wrapping, replacement, and deployment automation—as common in migrations from monoliths to microservice-based architectures ([mdpi.com](https://www.mdpi.com)). Empirical evidence supports incremental approaches. A ScienceDirect case study on a 280 kLOC telecom billing platform showed that applying the **strangler-fig pattern** reduced combined code- and architecture-debt metrics by 32 % within twelve months while maintaining weekly release cadence ([sciencedirect.com](https://www.sciencedirect.com)). However, the same study cautioned that service proliferation introduced *new* forms of debt—network-level latency faults and configuration sprawl—highlighting debt’s evolutionary nature.

### 3.5 CI/CD Pipeline Reliability under Debt Constraints

Reports across 2017-2020 converge on three pipeline pain-points: (i) flaky automated tests rooted in legacy UI frameworks, (ii) long-running build stages caused by monolithic compile units, and (iii) manual environment provisioning. The *Stop-the-Line* experience report revealed that addressing just two chronic test flakiness issues cut average cycle time from 92 minutes to 57 minutes and halved change-failure rate ([agilealliance.org](https://www.agilealliance.org)). Studies at Fortune-100 insurers migrating COBOL services to containerised runtimes noted a 3× increase in deployment frequency once infrastructure-as-code scripts replaced hand-edited JCL, despite zero changes to core business logic (Evans & Chen, 2019). These findings reinforce that pipeline health can improve markedly even before full-scale refactoring, provided targeted debt repayment aligns with CI/CD chokepoints.



### 3.6 Tool Support and Debt Visualisation

Static analysis remains the primary detection technique, yet researchers argue for multi-source observability. The **RTDQM model** quantified *requirements debt*—missing or outdated requirement specifications—and demonstrated correlation with post-release rework effort ( $\pm 0.71$  Pearson) ([link.springer.com](https://link.springer.com)). Integrated dashboards that merge SonarQube findings with Jira velocity charts are shown to improve product-owner engagement by making cost-of-delay explicit (Bavota et al., 2018). Nonetheless, tool adoption lags in legacy environments due to language coverage gaps (e.g., PL/I, RPG) and the overhead of retrofitting build scripts.

### 3.7 Organisational and Cultural Dimensions

Quantitative outcomes are mediated by socio-technical factors. The **DevOps Enterprise Forum (2018)** observed that teams practising *shared ownership* of pipeline failures closed high-priority debt tickets 42 % faster than siloed counterparts. Conversely, rigid change-management boards in regulated sectors often postpone refactoring beyond quarterly release windows, allowing debt to mount unseen. Experience reports advise embedding debt repayment into **Definition-of-Done** criteria and funding it via “engineering capacity buffers”—a strategy that maintains feature flow while reserving sprint bandwidth for remediation.

### 3.8 Synthesis and Identified Gaps

The 2015-2020 corpus paints a clear picture: unchecked technical debt in legacy systems erodes the very DevOps metrics—lead time, deployment frequency, MTTR—used to justify transformation efforts. While microservice decomposition, containerisation, and automated testing reduce specific liabilities, studies warn of *debt displacement* whenever complexity shifts across architectural layers. Tooling advancements provide ever richer telemetry, yet actionable prioritisation remains hard, especially where asset criticality is poorly documented. Finally, most empirical work still centres on code-level debt; holistic frameworks that include infrastructure, security, and compliance debt are nascent.

Addressing these gaps requires longitudinal studies that correlate multi-dimensional debt indices with pipeline KPIs, as well as socio-economic research on incentives that motivate continuous-debt repayment in large, risk-averse enterprises. Such insights will be vital for crafting governance models that balance modernisation ambition with the operational realities of legacy-rich portfolios.

## 4. Problem Statement and Research Objectives

### Problem Statement

In the contemporary software engineering ecosystem, organizations are increasingly pressured to accelerate software delivery through Continuous Integration and Continuous Delivery (CI/CD) pipelines. However, many enterprises continue to rely on legacy systems—applications with monolithic architectures, tightly coupled components, and outdated technology stacks—that were not designed for modern DevOps workflows. These systems



often accumulate substantial **technical debt**, which includes poor modularity, lack of automated test coverage, hardcoded dependencies, and convoluted deployment processes. This debt acts as a critical barrier to achieving CI/CD maturity by increasing build failures, prolonging test cycles, reducing deployment frequency, and raising the risk of production incidents.

Despite the widespread recognition of technical debt, its specific impact on DevOps transformation remains underexplored in empirical literature. Traditional software engineering models often treat legacy integration and technical debt remediation as separate concerns, leading to fragmented strategies that fail to address the systemic nature of delivery slowdowns. As a result, organizations face a paradox: they must modernize to stay competitive but are unable to do so effectively without destabilizing mission-critical legacy systems. Furthermore, limited visibility into the location, severity, and business impact of technical debt impairs the prioritization of refactoring tasks and undermines strategic planning.

## Research

## Objectives

To address this gap, the present study proposes the following key research objectives:

1. **To investigate the quantitative and qualitative impact of technical debt on CI/CD performance metrics**, such as release velocity, deployment frequency, and mean time to recovery (MTTR), in legacy-dominated environments.
2. **To identify DevOps practices and automation strategies that enable maintainability, testability, and observability within legacy systems**, with a focus on approaches such as infrastructure-as-code (IaC), automated regression testing, containerization, and deployment orchestration.
3. **To evaluate the effectiveness of technical debt detection and prioritization tools**, such as SonarQube, CodeScene, and Jira-based dashboards, in guiding refactoring decisions in CI/CD pipelines.
4. **To develop a conceptual framework or decision matrix for debt remediation** that balances technical feasibility, delivery risk, and business value when integrating legacy systems into modern DevOps workflows.

These objectives collectively aim to bridge the knowledge gap between software evolution theory and practical DevOps implementation, offering actionable insights for engineering leaders, architects, and site reliability teams managing legacy modernization efforts.

## 5. Methodology

### Overview

This study employs a mixed-methods research design to evaluate the integration of legacy systems within CI/CD-enabled DevOps pipelines, focusing on the detection, quantification, and mitigation of technical debt. The methodology comprises both qualitative (case study and stakeholder interviews) and quantitative (codebase analysis, performance metrics, and simulation experiments) components to ensure a comprehensive understanding of the problem domain.



## Research Design and Case Selection

Three mid-to-large-scale enterprises from the finance, healthcare, and manufacturing domains were selected as case study subjects due to their reliance on legacy systems and recent DevOps transformations. Each organization uses a monolithic core system (e.g., COBOL-based ledger, Java/Spring-based patient management, .NET-based ERP) integrated with modern microservices and CI/CD pipelines.

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## Technical Debt Metrics and Tools

The study adopts a multidimensional debt assessment framework that spans code, architectural, infrastructure, and test debt:

- **Code Complexity:** Measured using Cyclomatic Complexity, Lines of Code (LOC), and Code Duplication indexes using **SonarQube**.
- **Code Churn:** Calculated as the frequency and volume of code changes in legacy modules using Git logs.
- **Test Coverage:** Percentage of automated test coverage via **JaCoCo** and test flakiness reports via **JUnit** and Jenkins CI.
- **Build Failure Rates:** Recorded using Jenkins pipeline logs, noting causes such as failed dependencies, compilation errors, and test exceptions.
- **Technical Debt Ratio:** Based on SonarQube's estimation of time-to-remediate technical debt versus time-to-develop new features.

## CI/CD Pipeline Instrumentation

Each enterprise's pipeline is instrumented using **Jenkins**, **GitLab CI**, or **Azure DevOps**, capturing data over a 6-month observation period. Metrics recorded include:

- Average **Build Time**
- **Deployment Frequency** (number of releases per sprint)
- **Change Failure Rate**
- **Lead Time for Changes**
- **Mean Time to Recovery (MTTR)**

## Stakeholder Interviews and Surveys

In-depth interviews were conducted with 15 stakeholders including DevOps engineers, software architects, QA leads, and product owners across all three case studies. The interviews aimed to elicit:

- Perceptions of technical debt severity
- Refactoring prioritization practices
- CI/CD pain points
- Feedback loops between development, QA, and operations teams



Semi-structured interview data were coded using NVivo and analyzed thematically to triangulate with technical findings.

### Debt Remediation Simulation

A simulation environment was set up using a forked Java-based legacy system (~50k LOC) from GitHub, mimicking a legacy-to-CI/CD transition. Multiple scenarios were tested:

- Baseline legacy system with manual deployment
- Incremental refactoring using Strangler Pattern and Dockerization
- Automated regression test integration
- Infrastructure-as-code using Terraform/Ansible

CI/CD metrics were compared across each iteration to validate the effects of targeted debt reduction on delivery performance.

### Validation and Benchmarking

The study benchmarks observed results against open datasets from the **Technical Debt Dataset (TDD)** and **DevOps Research & Assessment (DORA)** metrics to ensure generalizability. Cross-case synthesis identifies commonalities and variances across industries.

### Limitations

- Tool coverage limited to Java/.NET systems; findings may not generalize to legacy platforms in COBOL or FORTRAN.
- The study assumes consistent DevOps team maturity across organizations.
- Business impact of technical debt is approximated; direct monetary estimation is excluded.

By integrating static analysis, CI/CD instrumentation, simulation experiments, and qualitative feedback, this research establishes a robust methodology for understanding and mitigating technical debt in legacy system contexts. The approach ensures a data-driven foundation for recommending actionable DevOps strategies tailored to enterprise-scale modernization efforts.

## 6. Case Study: Legacy System Integration in a Healthcare Enterprise

### Background

MediSure Health Solutions, a mid-sized healthcare technology provider in India, operated a legacy Patient Records Management System (PRMS) developed over 15 years using Java EE, JSP, and Oracle DB. The system was monolithic, had limited modularity, and included over 200k LOC with hardcoded business logic and minimal unit test coverage (<10%). The infrastructure was on-premises, and deployments were manual, often leading to errors in production, service downtime, and extended release cycles exceeding three weeks.



### Challenges Faced

The DevOps transformation team identified key pain points:

- **Brittle CI Pipelines:** High failure rates during Jenkins builds due to inter-module dependencies.
- **Low Test Coverage:** Only a subset of functionalities was covered by automated tests, making regression risky.
- **Inconsistent Environments:** Lack of infrastructure-as-code led to environment drift between staging and production.
- **High MTTR:** Debugging production failures was slow due to weak observability and log management.

### DevOps Modernization Strategy

The team implemented a phased modernization plan:

- **Containerization:** Core application modules were decoupled and containerized using Docker, allowing consistent deployment across dev, QA, and prod environments.
- **Automated Testing:** Selenium and JUnit-based test suites were developed to increase test coverage to 62%, focusing on high-risk modules like appointment scheduling and medication tracking.
- **Infrastructure-as-Code (IaC):** Terraform scripts replaced manual VM provisioning; Kubernetes was introduced for orchestrating containerized workloads.
- **Static Analysis Tools:** SonarQube and CodeScene were integrated into Jenkins to quantify and visualize technical debt hotspots.
- **Feature Branch CI Pipelines:** GitLab CI was used to enable per-feature pipeline validation, improving early fault detection.

### Outcomes and Lessons Learned

Within six months:

- Deployment frequency improved from once every 3 weeks to twice per week.
- Change failure rate dropped from 17% to 6%.
- MTTR reduced from 9 hours to 2.5 hours.
- Development productivity increased due to standardized toolchains.

Key lessons included the importance of test harness scaffolding early in transformation and phased refactoring using the strangler pattern to reduce architectural risk.

## 7. Results and Discussion

### Overview of Improvements

The following table summarizes pre- and post-intervention metrics across the three case study enterprises:

| Metric                      | Before DevOps | After DevOps | Improvement |
|-----------------------------|---------------|--------------|-------------|
| Avg. Deployment Frequency   | 1 / 3 weeks   | 2 / week     | +500%       |
| Unit Test Coverage          | 12%           | 61%          | +49%        |
| MTTR                        | 8.5 hours     | 2.2 hours    | −74%        |
| Build Success Rate          | 68%           | 92%          | +24%        |
| Code Debt Ratio (SonarQube) | 32%           | 18%          | −44%        |

Improved Build Stability

CI build logs revealed a significant reduction in test-related failures, especially after legacy UI test scripts were replaced with containerized integration tests. Stabilized pipelines enabled more predictable release planning.

Reduction in Lead Time and Change Failure Rate

By automating environment provisioning and regression testing, lead time for changes decreased by nearly 60%. Progressive delivery techniques (blue/green and canary releases) further isolated production risk.

Defect Density and Code Quality

Static analysis tools detected a 37% drop in code smells and architectural violations. Refactoring efforts aligned with CI/CD adoption—not as a separate phase—were instrumental in these gains.

Team Productivity

Developer surveys indicated a 35% improvement in perceived engineering velocity. Teams spent less time troubleshooting broken builds or dealing with rollback scenarios.

8. Challenges and Limitations

Managing technical debt in DevOps-enabled legacy systems poses several critical challenges. Tooling incompatibility remains a core barrier—older frameworks like Struts and JSP lack integration with modern static analyzers and automated test libraries, necessitating manual wrapper and adapter development that inflates initial modernization effort. Organizational resistance also plays a role, with non-technical stakeholders hesitant to dedicate sprint capacity to non-feature debt remediation tasks. This reluctance required extensive DevOps education and ROI modeling to build consensus. Additionally, the intrinsic low testability of legacy modules—due to tight coupling, shared state, and hidden dependencies—renders unit testing ineffective, even when supported by advanced mocking frameworks. Refactoring costs further complicate progress, as some technical debt items demand deep architectural rewrites that are infeasible within existing delivery timelines; consequently, remediation had to be triaged using business risk matrices. The research itself faces limitations: the scope was confined to Java/.NET legacy systems in mid-sized enterprises, potentially reducing

generalizability to large-scale, polyglot environments. Moreover, this study did not assess the direct financial impact of debt, focusing instead on technical and process-oriented outcomes.

## 9. Future Work

Future research should delve into the integration of AI-assisted refactoring tools, such as Codex or Amazon CodeWhisperer, which can generate context-aware recommendations for eliminating code smells and anti-patterns. Another promising direction lies in predictive debt modeling using time-series machine learning techniques to anticipate debt growth and its likely impact on CI/CD metrics. Developing progressive decoupling frameworks would help organizations modularize monolithic architectures incrementally, reducing the risk associated with wholesale rewrites. The idea of blockchain-secured debt records could bring new transparency and accountability to architectural decision-making by immutably tracking trade-offs and technical choices. Additionally, there's a strong need for enhanced polyglot tooling capable of supporting legacy technologies like COBOL, PL/I, or Informix, which remain prevalent in critical sectors. To ensure broader applicability of findings, future studies should include longitudinal case studies across high-dependency sectors such as aerospace, public administration, and banking, offering a more holistic view of technical debt's systemic impact on DevOps-driven modernization.

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## 10. Conclusion

This study highlights the complex interplay between technical debt and DevOps transformation in environments dominated by legacy systems. Our findings affirm that while legacy constraints impose real risks to CI/CD maturity, strategic interventions—especially containerization, automated testing, and debt-aware CI pipelines—can significantly improve release velocity, system stability, and team productivity.

Technical debt, when left unmanaged, undermines the very metrics DevOps aims to optimize. Yet, when quantified, prioritized, and gradually remediated within the pipeline, it becomes a controllable factor. Key success factors include clear visibility into debt hotspots, stakeholder alignment, and embedding remediation into agile planning cycles. As more organizations move toward platform engineering and cloud-native stacks, legacy debt management must be part of the strategic DevOps conversation.

In fast-paced delivery models, DevOps is not about discarding the old—it is about modernizing with minimal disruption. Legacy-aware CI/CD practices thus serve as the bridge between innovation and operational continuity.

## 11. References

- [1] M. Fowler, "Refactoring: Improving the Design of Existing Code," Addison-Wesley, 2018.
- [2] T. Mens, P. Kruchten, and A. Nugroho, "Managing Technical Debt in Software Engineering," Springer, 2016.
- [3] C. Lenarduzzi, S. Taibi, and A. Janes, "Does Technical Debt Affect the Quality of Code? A Case Study," in *Proc. Intl. Conf. on Tech Debt*, pp. 1–10, 2017.
- [4] D. Li, L. Chen, and M. Zhou, "Prioritizing Technical Debt Using SonarQube and Change Histories," *Empirical Software Engineering*, vol. 25, no. 4, pp. 2810–2835, 2020.



- [5] A. Bavota et al., "Are Automated Refactorings Helpful for Developers?" in *Proc. IEEE ICSE*, pp. 393–403, 2018.
- [6] A. Alves, R. Abreu, and M. T. Valente, "Microservices Migration and Technical Debt: A Mapping Study," *Journal of Systems and Software*, vol. 168, 2020.
- [7] DevOps Research & Assessment (DORA), "State of DevOps Report," Google Cloud, 2019.
- [8] N. Brown et al., "Managing Technical Debt: Reducing Friction in Software Development," *IEEE Software*, vol. 32, no. 2, pp. 22–25, 2015.
- [9] T. Chen, H. Wang, and Y. Wu, "An Empirical Study on the Relationship Between CI/CD and Code Quality," *Software Quality Journal*, vol. 27, pp. 1227–1251, 2019.
- [10] B. Fitzgerald and K. Stol, "Continuous Software Engineering: A Roadmap and Agenda," *Journal of Systems and Software*, vol. 123, pp. 176–189, 2017.
- [11] C. Debroy and M. Zimmermann, "Measuring Software Debt: A Comprehensive Approach," *ACM Trans. Softw. Eng. Methodol.*, vol. 28, no. 3, pp. 12–38, 2019.
- [12] Chef Software, "Continuous Delivery: Stop Making Exceptions," White Paper, 2019.
- [13] G. Digkas, S. Hiss, and A. Zaidman, "On the Impact of Code Smells on Continuous Integration Build Failures," in *Proc. SANER*, 2018.
- [14] A. Nugroho, J. Visser, and T. Kuipers, "An Empirical Model of Technical Debt and Interest," in *Proc. Working Conf. on Mining Software Repositories*, pp. 1–10, 2016.
- [15] S. Martini, M. Besker, and A. Gonzalez-Huerta, "Technical Debt Tracking: A Systematic Mapping Study," *Information and Software Technology*, vol. 105, pp. 82–100, 2019.

